

Network optimization tools

	Planning	Tuning	Troubleshooting
NWS	NWS-A	NWS-A	NWS-A
R-PMO		R-PMO	R-PMO
RNO	TET	NCS/NOX FAS/FOX TET MRR	NCS/NOX FAS/FOX MRR
PMR			MTR CTR CER

Radio Network Optimization Tools in GSM

RNO Applications in GSM OSS

- Frequency Allocation Support (FAS) and Frequency Optimization Expert (FOX)
- Neighbor Cell Support (NCS) and Neighboring Cell Optimization Expert (NOX)
- Measurement Result Recordings (MRR)
- Traffic Estimation Tool (TET)

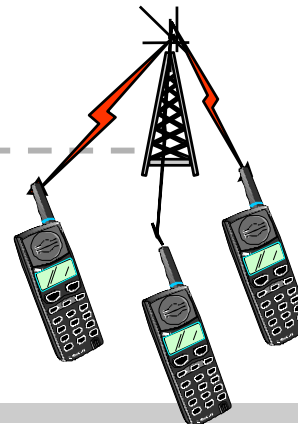
Required
subsystems in BSC

MRR

BAR

RIR

RNO uses measurement
from all active mobiles
in the network on
both uplink and downlink



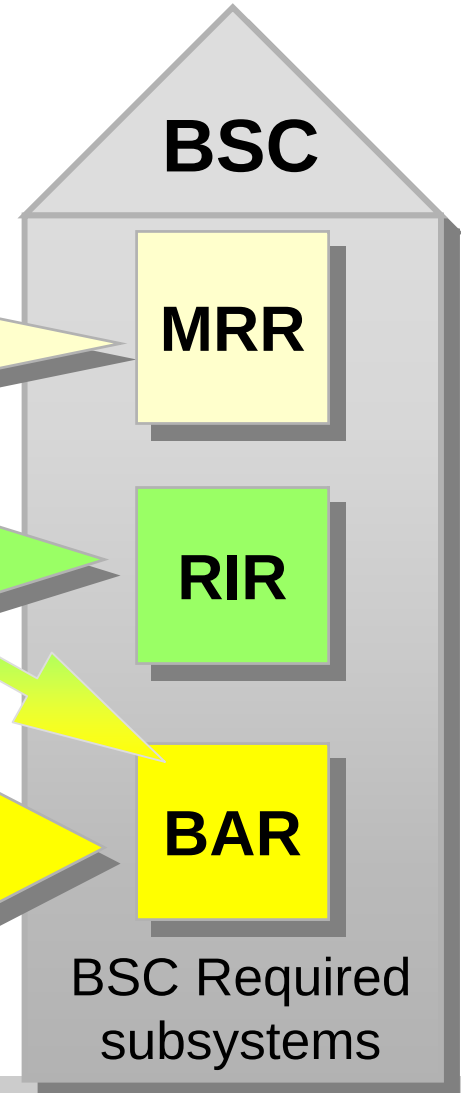
RNO - Radio Network Optimization

Applications

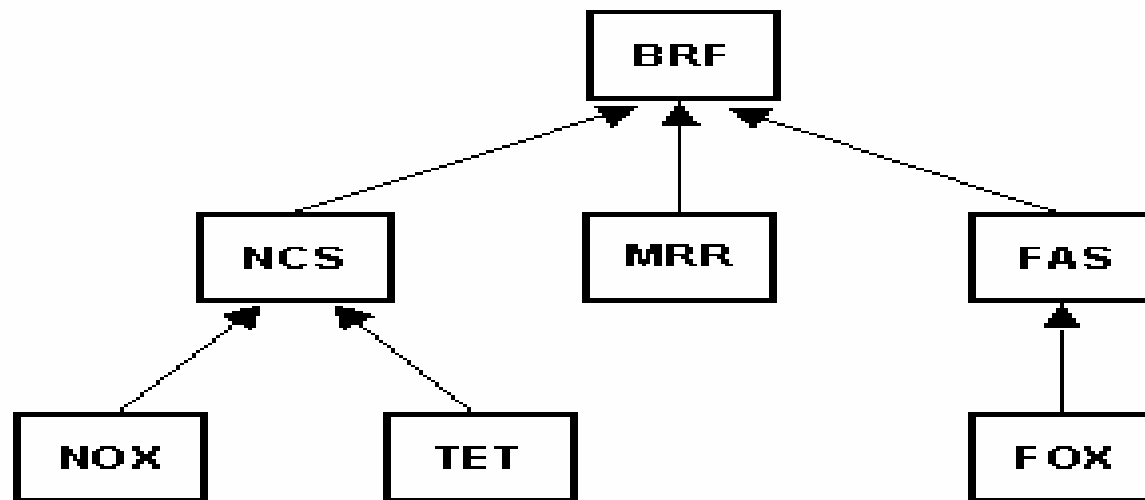
MRR - Measurement Result Recording

FAS - Frequency Allocation Support
FOX - Frequency Optimization eXpert

NCS - Neighbor Cell Support (NCS)
NOX - Neighboring cell Optimization eXpert

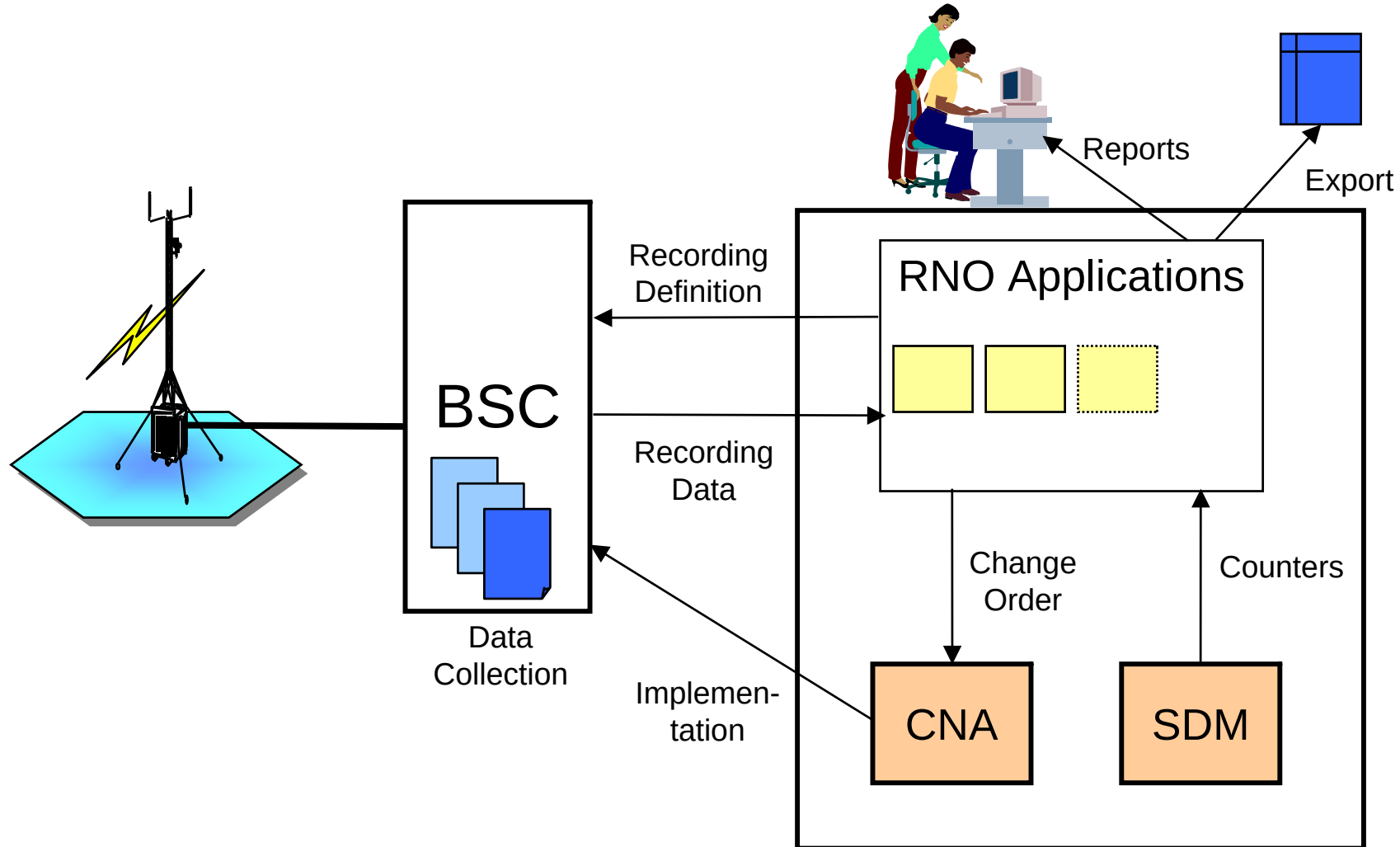


RNO Installation



**BRF=Basic Recording
Framework for RNO products**

RNO - Common Structure



RNO System Parameters

- Most of the RNO PDB parameters have default values which is good enough to start working with, but some must be edited before you can start running RNO:
 - **Band Parameters**
 - **BCCH Frequency Numbers**
 - **STS Collection Timeouts**
- Some of the PDB parameters can be edited in a dialogue box of the RNO application. The dialogue box is opened by selecting the RNO System Parameters from the Options menu in the RNO main window (see the on-line help).
- The function of FAS, NCS, FOX and NOX depends on information about the **geographical position, antenna type, and direction of cells**. This information is used when BCCH/BSIC combinations from the BARFIL are translated into the corresponding cell names. The geographical position data is also used when RNO recording data is shown in the GNIP tool.

- RNO reads the position data from CNA.
- Some of the data (which RNO needs) is not read into CNA when doing a CNA-adjust
- You need to install SMIA, SGW and SDM to get STS counters into RNO.
- The optional features RADIOINTREC and MEASRESREC must have been activated in the BSC in order to run FAS/FOX and MRR.
- The status of the recording result will vary from "Processing" to "Ready for Analysis" until the STS data arrives. **This may be up to 4 hours after the end of the recording.**

The following CNA attributes are not included in CNA-adjust

Object	Attribute	Value Range
site	latitude	Latitude-milliseconds
site	longitude	Longitude-milliseconds
cell	antenna_type	("OMNI", "SECTOR" or "ADAPTIVE")
cell	cell_dir	0..360 (Integer)
cell	latitude	Latitude-milliseconds
cell	longitude	Longitude-milliseconds

RNO - System Parameters

RNO – System Parameters

Application: RNO

Parameter	Value
unit	dBm
GSM800Band	
GSM900Band	1-26
GSM1800Band	520-532
GSM1900Band	
BCCHFreqNos	1-26, 532
blockEdge	
coFreqDist	20
adjFreqDist	10
freqsAdded	5
pathLossPerDistDecade	31
sectorAntennaGainOffset	-5
sectorAntennaGainAmplitude	14
weakPCNChoiceWarningThreshold	5
BAlistChangeInterval	10

New Value:

Save

Description

GSM 800 Frequency Numbers

This is a list of frequency number ranges in the GSM 800 band, for which the operator has got a licence. The parameter is used for recording configuration data, and to define on which bands the set handling shall be supported.

Note: Changing this parameter will affect the next recording period after the change for all ongoing RNO recordings.

Type: String

Syntax: Frequency intervals (from-to) or single frequencies, comma separated.

Value range: 128-251

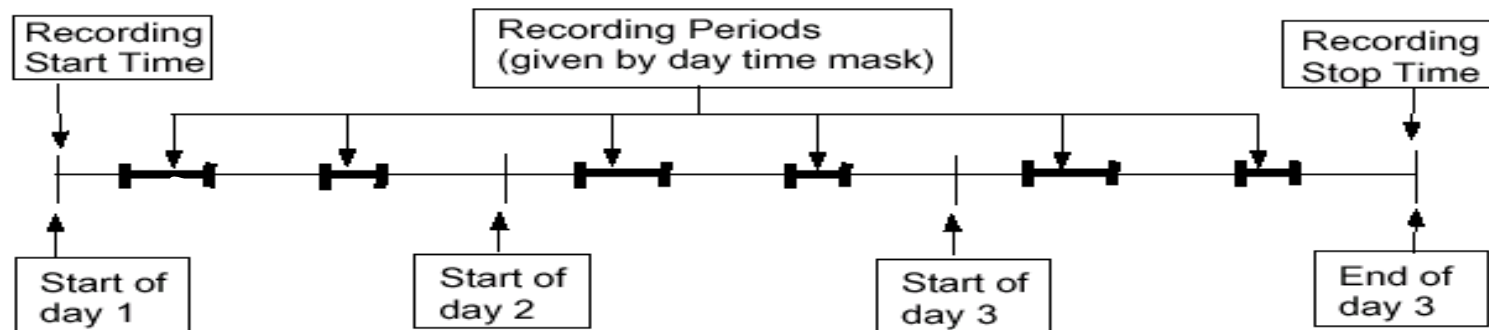
Example: 128-140,232-248,250

Installation setting:

Used by: FAS, FOX, NCS, NOX, TET, MRR

Close Help

Recording Session



Time

Start Date: 2002-06-12

Repeat: Weekly 1 Times

Days: Free selection ☐ Sun ☐ Mon ☐ Tue ☐ Wed ☐ Thu ☐ Fri ☐ Sat

Hours: - - - -

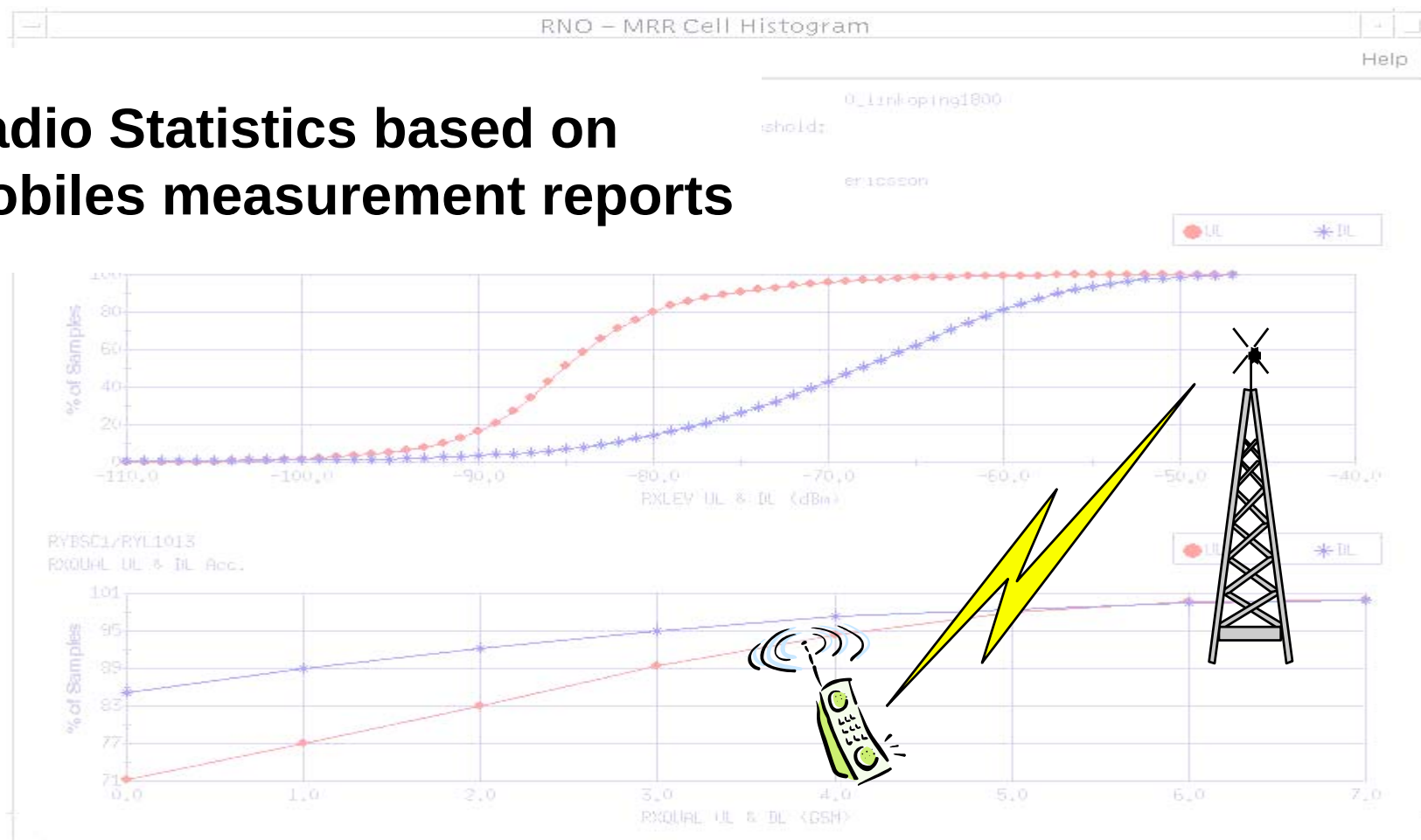
Start Date Only
Daily
Weekly
Every Two Weeks

Free selection
Workingdays
Weekend
Free selection

MRR

MRR - Measurement Result Recording

Radio Statistics based on Mobiles measurement reports



MRR collects data from Measurement Results

- UL RxLev (0,...,63 GSM or -110,...,-47dBm).
- DL RxLev (0,...,63 GSM or -110,...,-47dBm).
- UL RxQual (0,...,7 GSM).
- DL RxQual (0,...,7 GSM).
- TA (0,...,219 GSM).
- MS Power level (0,...,39 dBm).
- BTS Power reduction (0,-2,...,-28,-30 dB).

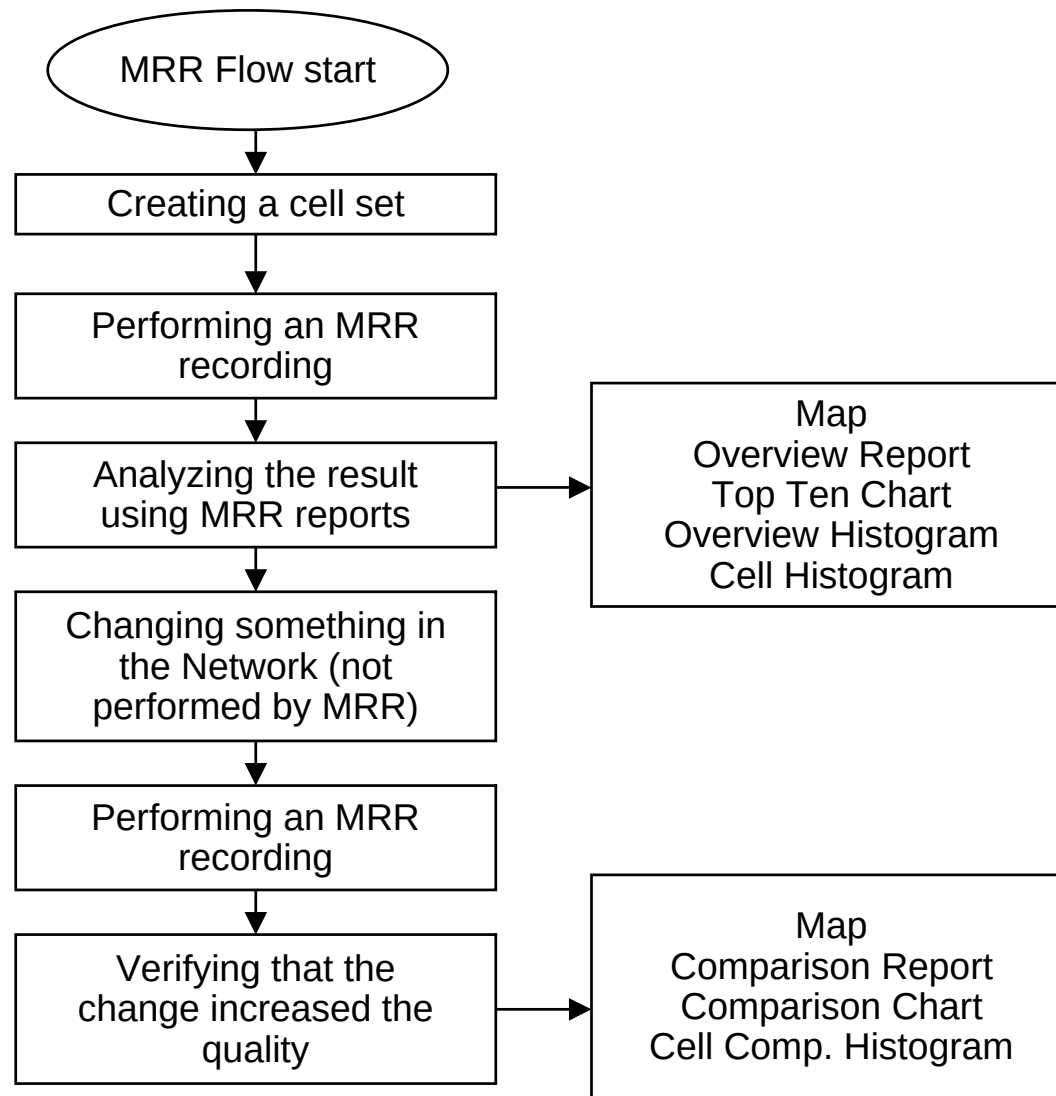
MRR separately calculates and accumulates counters for two other radio characteristics

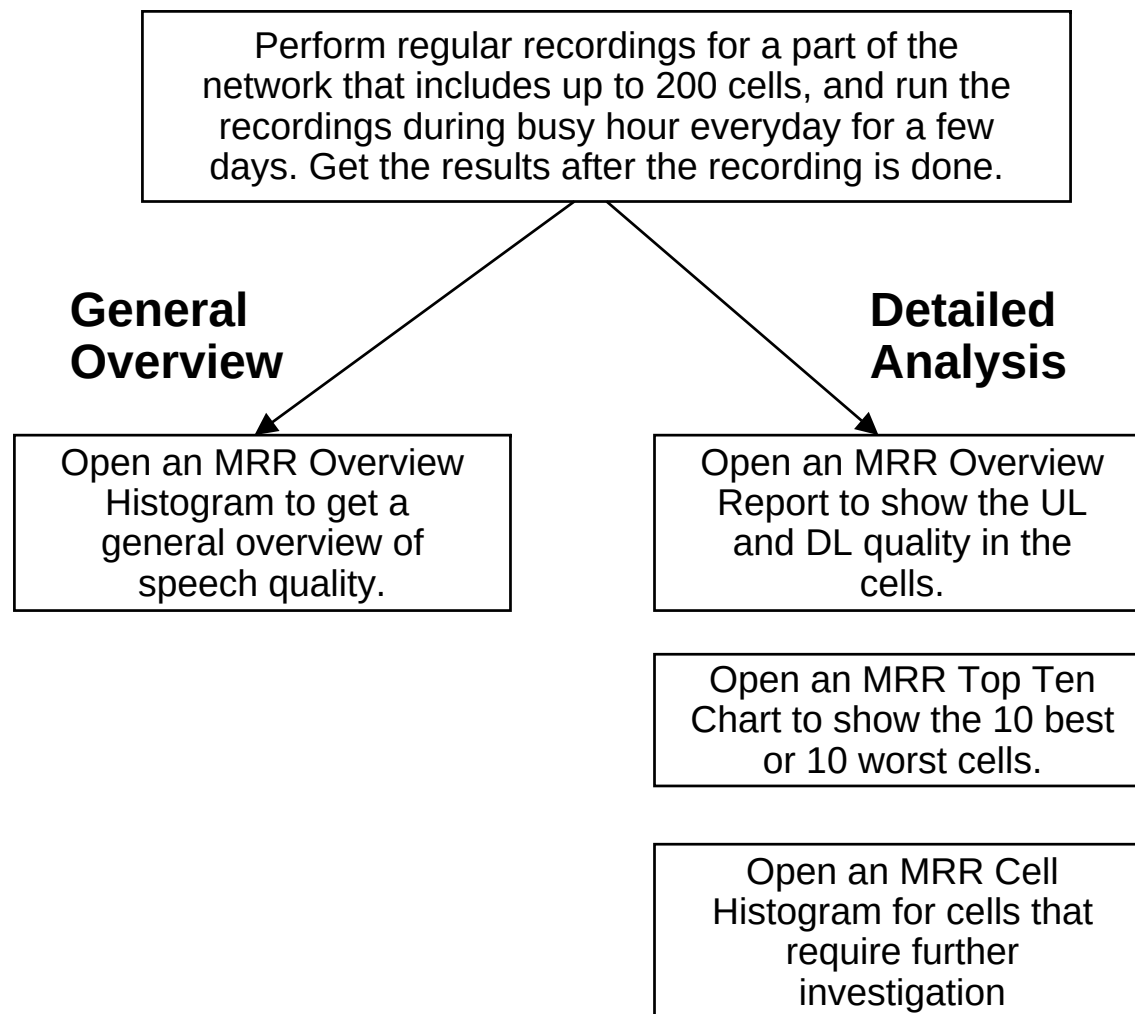
- UL Pathloss in dB (30,...,153 dB)
- DL Pathloss in dB(30,...,190 dB)

MRR - Example of usage

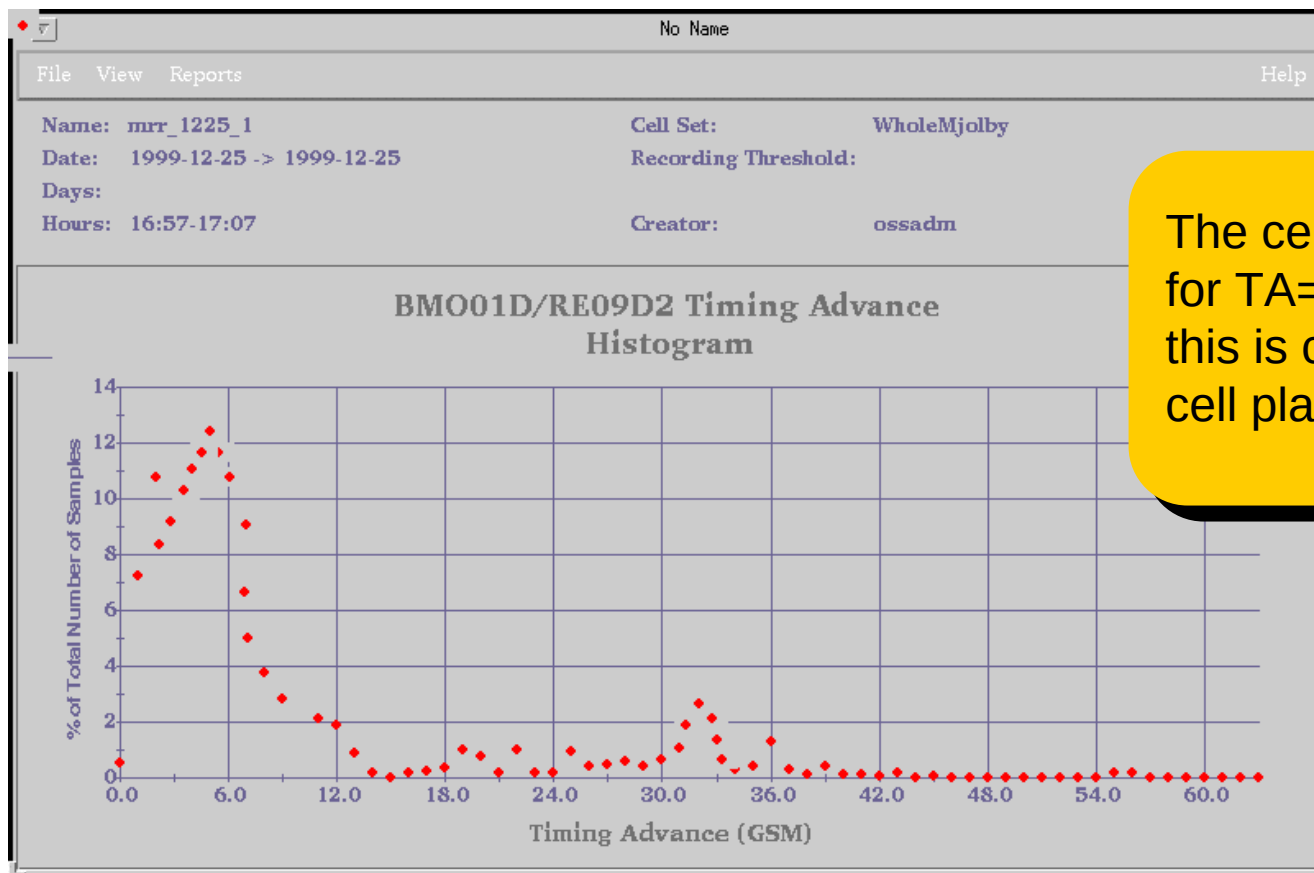
- Bad speech quality
- Interference detection
- Tuning of HCS parameters
- Cell covers to much







Cell covers too much (TA - recording)



The cell carries traffic for TA=32. Check if this is consistent with cell plan (map)

MRR - Important Recording Parameters

Get Result

☒ After Each Period, Accumulate Results
☐ After Each Period, Save Result Separately

Cell Set

Cell Filter: GSM900 ▼

West
0700
0700005 (GSM900)
0700006 (GSM900)
0700007 (GSM900)
0700008 (GSM900)

Select Cell Set... Select Cells...

Recording Threshold

☒ Record All Measurement Reports
☐ Record Only Measurement Reports with:

RXQUAL UL = GSM AND RXLEV UL = dBm

RXQUAL AND RXLEV ▼ UL ▼ = ▼

MRR - Beneficial Features

- Threshold measurements
 - Record measurements with RxQual > 4
- Correlation measurements with two thresholds
 - Signal Strength Vs Signal Quality
 - Signal Strength Vs Timing Advance
 - Signal Quality Vs Timing Advance

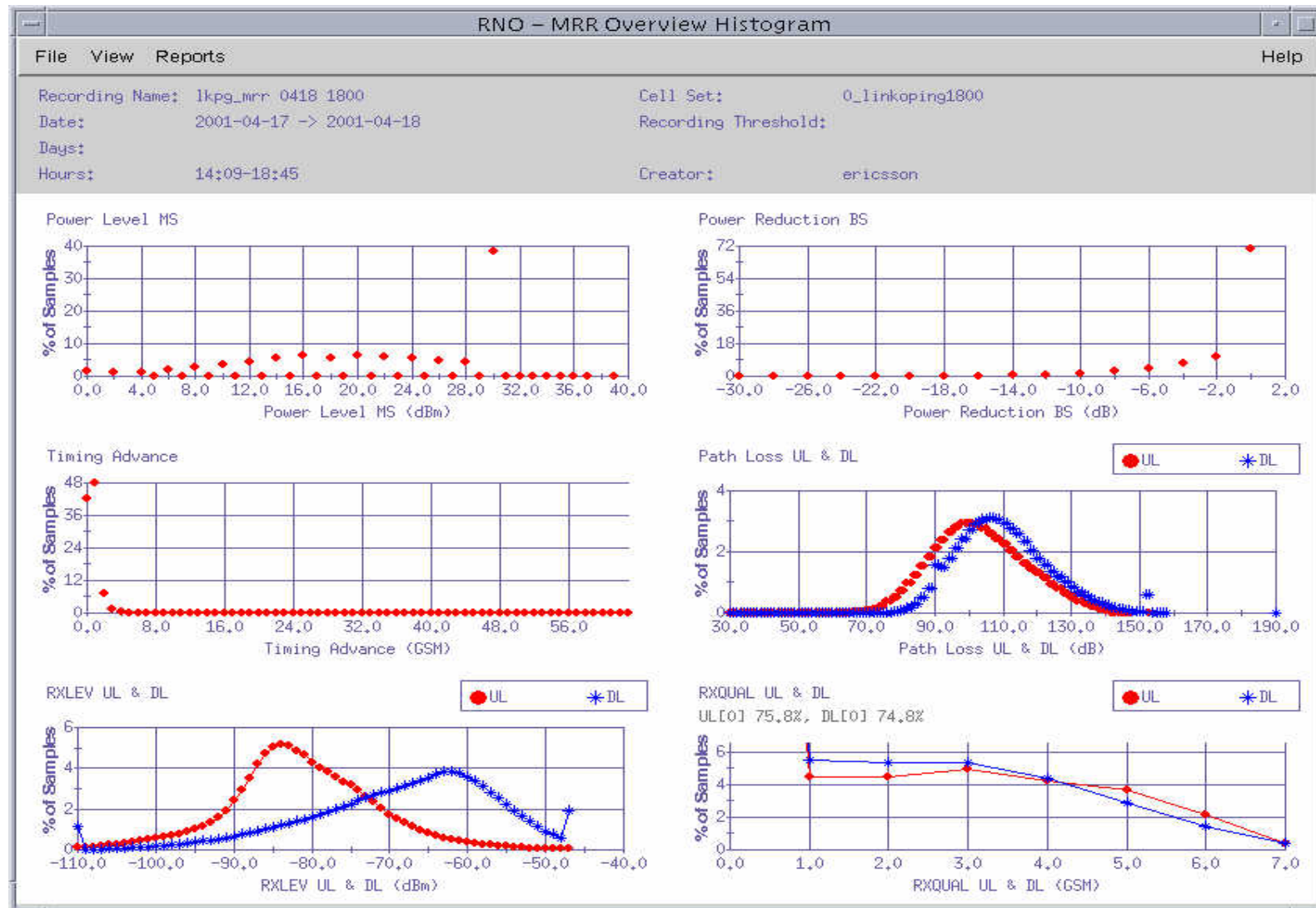
Example

- Find at which distance from the BTS interference occurs, e.g. using thresholds RxLev_DL > -85 dBm and RxQual_DL > 5

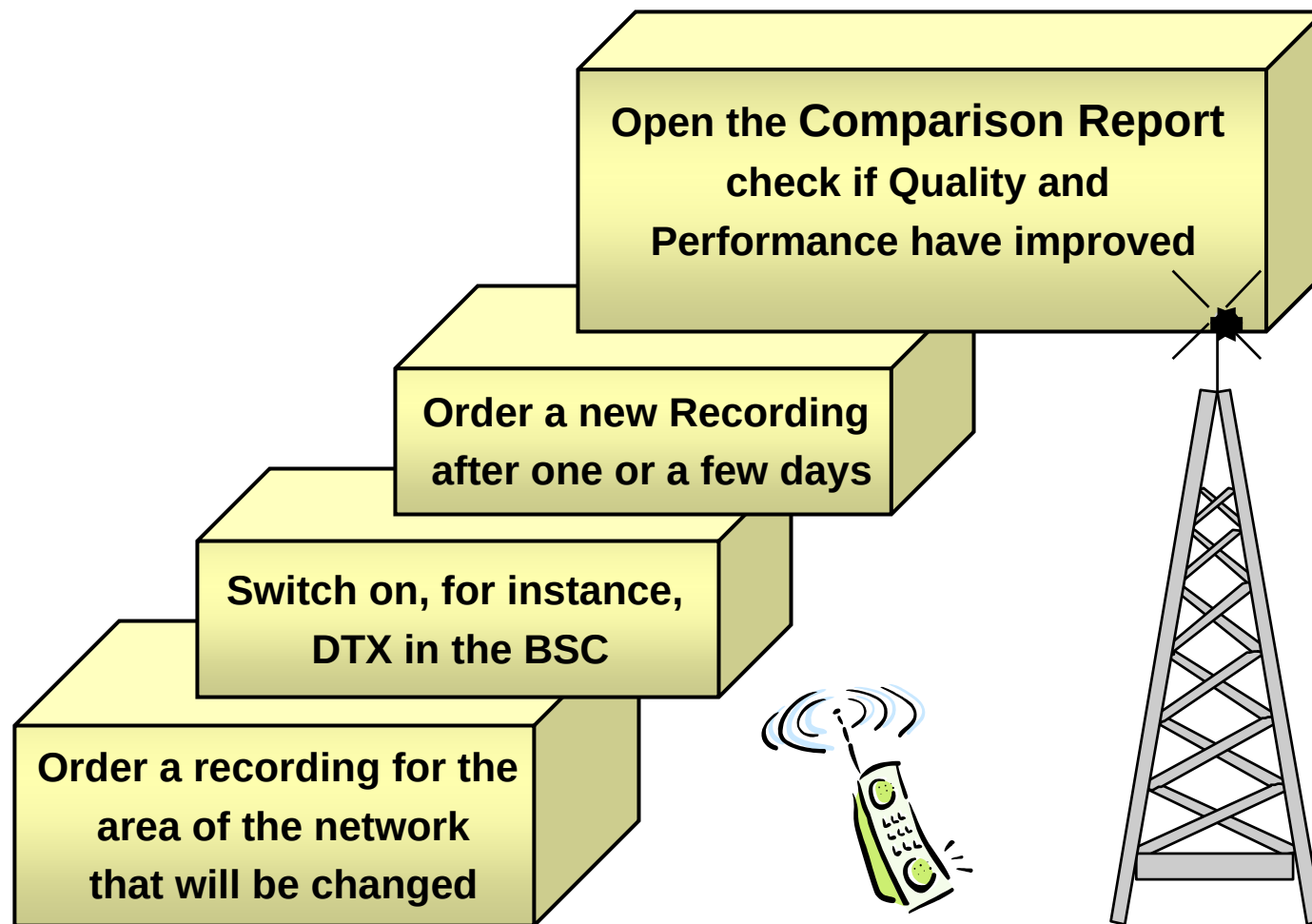
MRR - Reports

- Overview Report
- Overview Histogram
- Cell Histogram
- Trend Report
- Top Ten Chart
- Comparison Report
- Cell Comparison Histogram
- Comparison Chart

MRR - Overview Histogram



Finding the Effects of a Change



Radio Network Optimization (RNO)

File Edit View Options Reports Tools

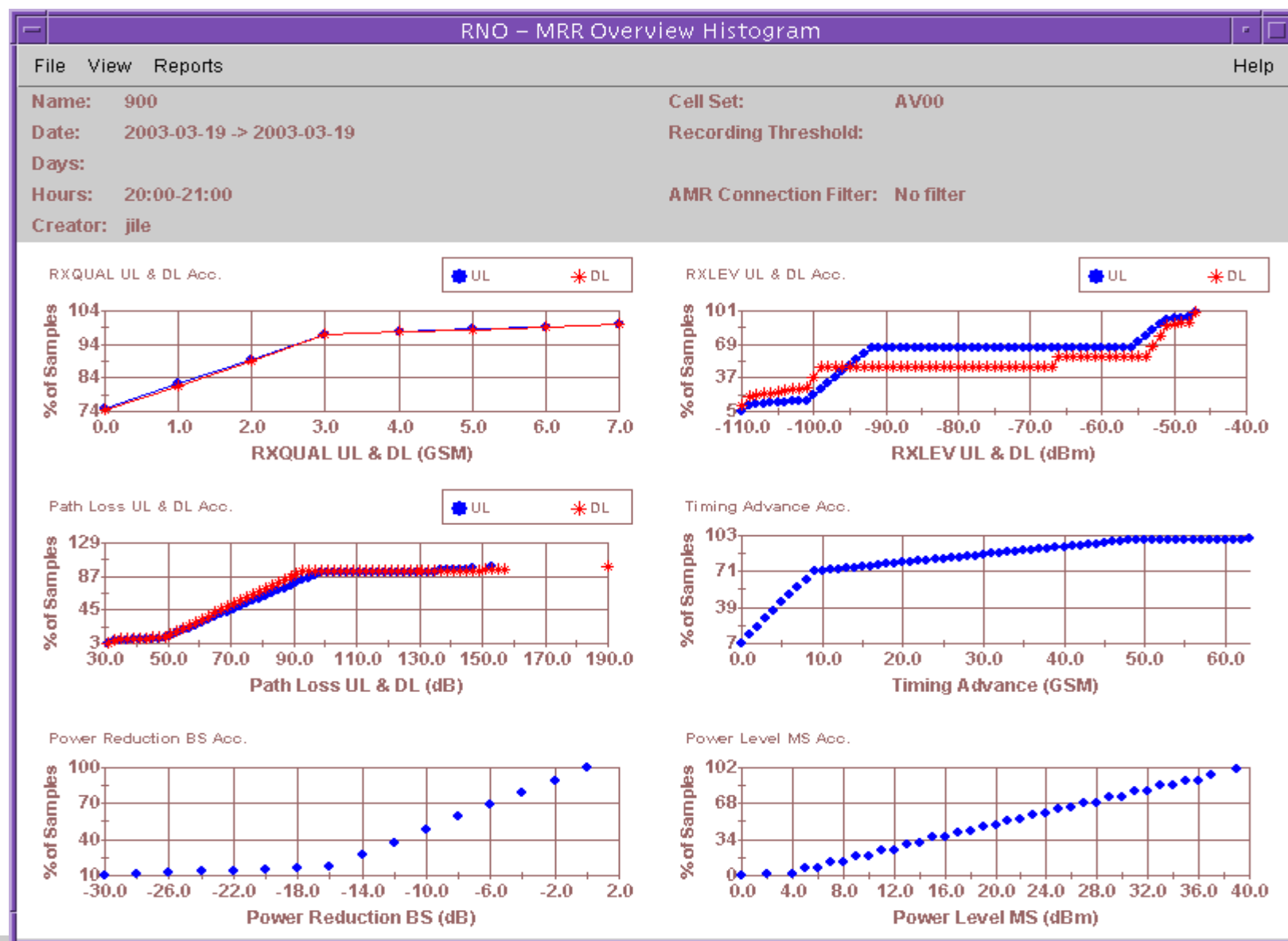
Help

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Recording Activities Recording Results

Recording Name	Type	Status	Cell Set	Cell Filter	# Cells	Creator	Start	Stop	# Rec. Periods
mrr1800	MRR	Completed #	AV00	GSM1800	30	jile	2003-03-19 11:04	2003-03-19 11:06	1 (1)
noxUpdate	NOX Aut.	Completed #	AV00	GSM900	55	jile	2003-03-12 09:51	2003-03-12 10:05	1 (1)
rirSts	FAS	Completed #	AV00	GSM900	55	jile	2003-03-19 16:31	2003-03-19 16:33	1 (1)
R3	NCS	Completed #	5000_900_R3	GSM900	60	jile	2002-03-27 10:25	2002-03-27 10:40	1 (1)
nox	NOX Recorn.	Completed #	AU00	GSM800	40	jile	2003-03-10 15:06	2003-03-10 15:23	1 (1)
largeMrr	MRR	Completed #	R10	-	2100	jile	2003-03-13 17:30	2003-03-13 23:00	4 (4)
largeFas	FAS	Completed #	R10_all_900	GSM900	495	bjek	2003-03-17 10:40	2003-03-17 11:55	0 (3)
fasBsm3	FAS	Completed	AV00_900Inbcch	GSM900	40	jile	2003-03-12 14:14	2003-03-12 14:16	1 (1)
fasBsm2	FAS	Completed	AV00_900Inbcch	GSM900	40	jile	2003-03-12 13:29	2003-03-12 13:31	1 (1)
mrrNonAmr	MRR	Completed	AU	-	200	jile	2003-03-13 10:44	2003-03-13 10:46	1 (1)
rirSts	FAS	Completed	AV00	GSM900	55	jile	2003-03-19 17:27	2003-03-19 17:29	1 (1)
fasSts	FAS	Completed	AV01	GSM900	55	jile	2003-03-21 09:34	2003-03-21 09:36	1 (1)
mrrOnlyAmr	MRR	Completed	AU	-	200	jile	2003-03-13 11:11	2003-03-13 11:13	1 (1)
21	MRR	Completed	C100	-	100	jile	2003-03-10 13:55	2003-03-10 13:57	1 (1)
R3	TET	Completed	5000_900_R3	GSM900	60	jile	2002-03-27 10:50	2002-03-27 12:15	4 (4)

Recording Comments



RNO – MRR Report Preferences

Comp. Chart Overv. Hist. Cell Hist. Cell Comp. Hist. Trend Report Stat. Calculations

Overv. Report Top Ten Chart Comp. Report

Report Columns

Available Columns

- No. of Meas.Rep Passed Filter
- Traffic Level Min/Max
- RXQUAL UL > 4
- RXQUAL UL Percentile 90
- RXQUAL DL > 4
- RXQUAL DL Percentile 90
- RXLEV UL > -95
- RXLEV UL Percentile 90
- RXLEV DL > -95

Included in Report

- Band
- Cell Type
- Channel Group
- Meas.Rep. Passed Filter (%)
- RXQUAL UL Average
- RXQUAL DL Average
- RXLEV UL Average
- RXLEV DL Average
- Traffic Level Average

Sort

Column: Cell Name

Order:

- ☒ Ascending
- ☐ Descending
- ☒ Keep Rows From Same Cell Together

Data Resolution

- ☒ Cell / Subcell
- ☐ Channel Group

Filter

☒ Off

☐ On

Show rows with: Cell Name =

OK Apply Cancel Help

RNO – MRR Report Preferences

Comp. Chart Overv. Hist. Cell Hist. Cell Comp. Hist. Trend Report Stat. Calculations

Overv. Report Top Ten Chart Comp. Report

		Statistical Threshold 1:		Statistical Threshold 2:	Percentile (%):
RXQUAL UL (GSM):	> ▼	4	- ▼		90
RXQUAL DL (GSM):	> ▼	4	- ▼		90
RXLEV UL (dBm):	> ▼	-95	- ▼		90
RXLEV DL (dBm):	> ▼	-95	- ▼		90
Path Loss UL (dB):	> ▼	145	- ▼		90
Path Loss DL (dB):	> ▼	150	- ▼		90
Path Loss Diff. (dB):	> ▼	0	- ▼		50
Timing Adv. (GSM):	> ▼	55	- ▼		90
Power Red. BS (dB):	= ▼	0	- ▼		90
Power Lev. MS (dBm):	= ▼	33	- ▼		90

OK Apply Cancel Help

This tab is used to define statistical calculations for the radio characteristics in the reports

The list of Recording Activities contains the following data per recording:

Name

Type

Recording status

Cell Set

Cell filter

Number of cells in the recording

Creator

Start time

Stop time

Number of retrieved and ordered recording periods.

Summary

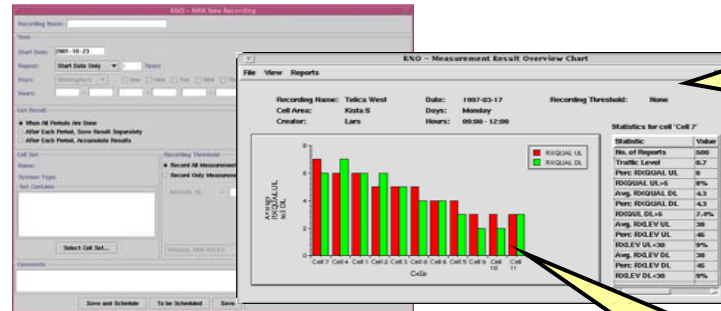
MRR - Summary

Collects and presents radio environment data from all mobiles (UL and DL) not available in any other tool

Recording threshold:

- RXQUAL(UL/DL)
- RXLEV(UL/DL)
- Timing Advance

Start recording



Reports

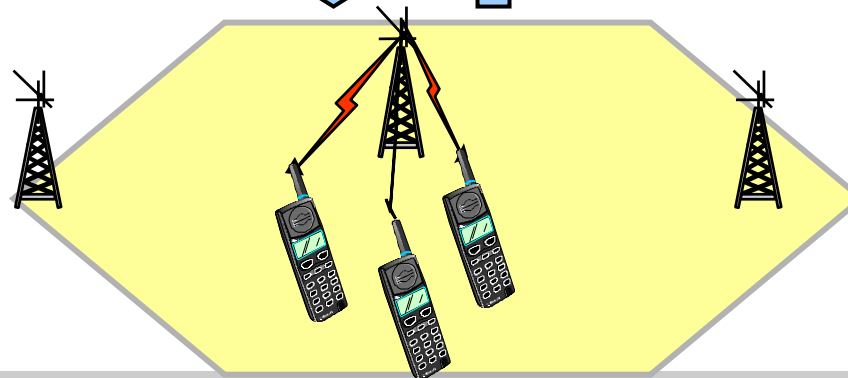
- Overview reports
- radio characteristics histograms
- comparison reports
- trend reports

Radio Characteristics:

- RXQUAL UL & DL
- RXLEV UL & DL
- Path loss UL & DL
- Timing Advance
- Power level, MS
- Power reduction, BTS

MRR

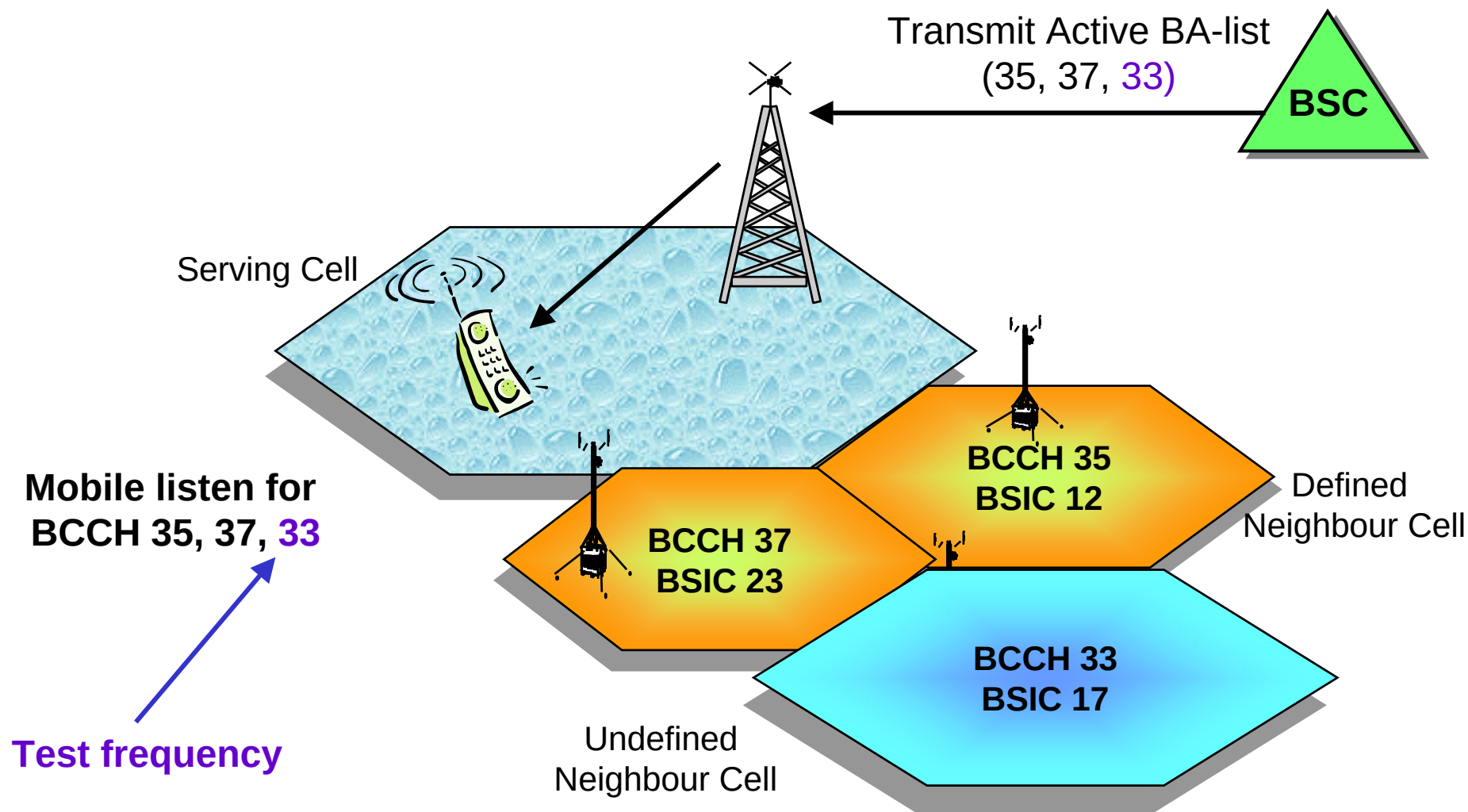
Measurement Reports



BAR

- The BSC Recording function active BA list Recording (BAR) adds BCCH-carriers, to the active BA list in a group of cells selected by the user.
- BAR then records, among other things, the average signal strengths on each of these frequencies with BSICs.
- The data is taken from Measurement Reports sent by active mobiles in the measuring cells.

Active BA-list Recording



RIR

- The BSC recording function Radio Interference Recording (RIR) performs Recordings of the uplink interference.
- During a Recording each TRX (within the Recording Cell Set) measures the uplink interference for the frequencies specified by the user.